DIFFERENTIAL BACKLASH MEASUREMENT/ADJUSTMENT

id051700665400

Preparation Before Servicing

1. Print out the measurement/adjustment value input sheet. (See MEASUREMENT/ADJUSTMENT VALUE INPUT SHEET.)

Note

- When performing the measurement/adjustment, input the measured and calculated values into the measurement/adjustment value input sheet.
- If the measurement/adjustment value input sheet has already been printed out for the other measurements/adjustments, use the sheet.

Differential Backlash Measurement

Note

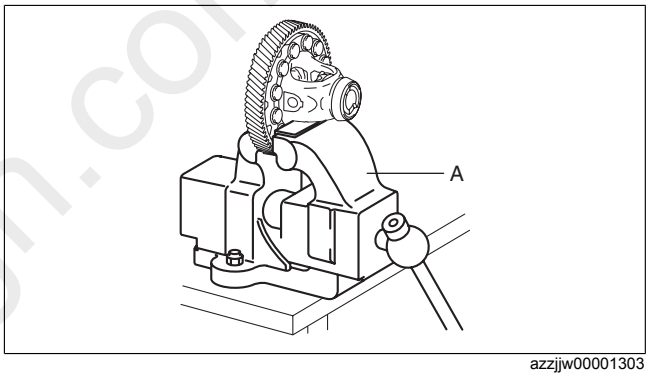
- Measure the tooth play (gap) between the side gear and pinion gear as the differential backlash.

1. Secure the ring gear and differential in a vise.

Caution

- **Insert a protective plate between the vise and the part so as not to damage the part.**

A : Vise



azzjjw00001303

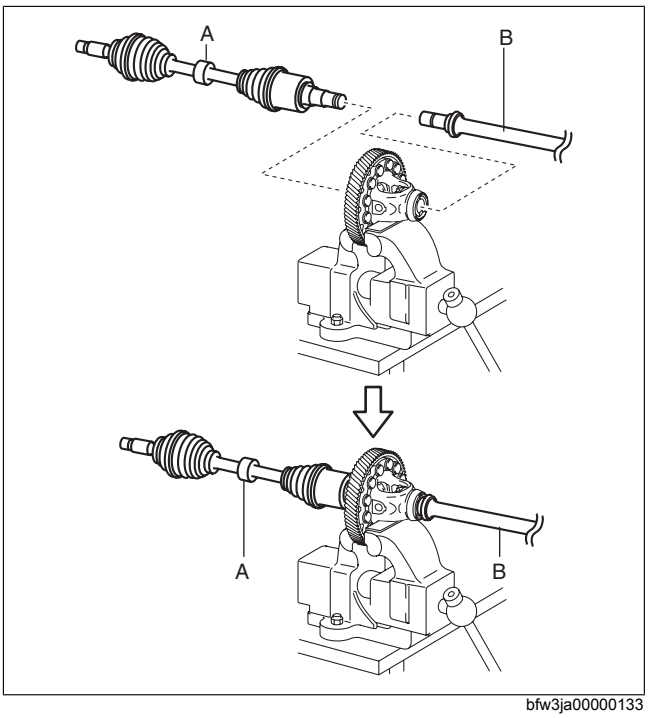
2. Assemble the drive shaft to the ring gear and differential.

Caution

- **Because the drive shaft (LH) clip is not required for the differential backlash measurement, do not assemble it.**

A : Drive shaft (LH)

B : Drive shaft (RH)



bfw3ja00000133

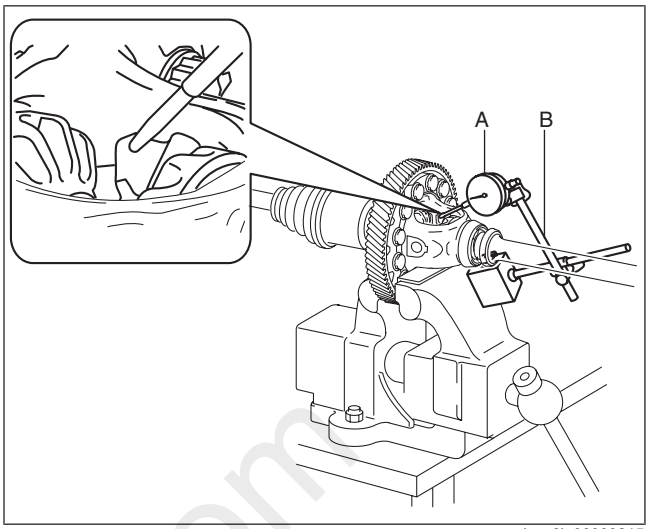
cardiagn.com

3. Set the dial gauge and magnetic stand as shown in the figure.

Caution
- **To reduce difference during the backlash measurement, set the dial gauge so that it is perpendicular to the teeth of the pinion gear.**

A : Dial gauge

B : Magnetic stand

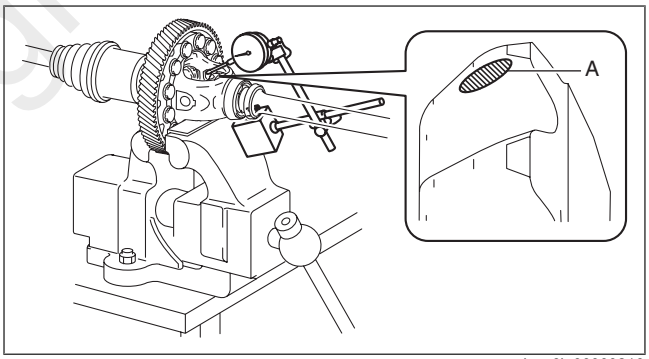


bcw6ja00000215

4. Set the dial gauge end to the pinion gear teeth.

Caution
- **To reduce difference during the backlash measurement, set the dial gauge end within the area shown in the figure.**

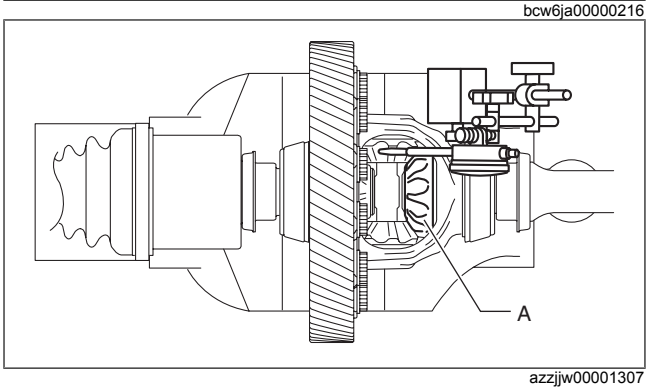
A : Dial gauge end set area



bcw6ja00000216

5. Secure the side gear on the front side by hand.

A : Secure by hand



azzjjw00001307

6. Move the pinion gear with the dial gauge by hand and measure the backlash.

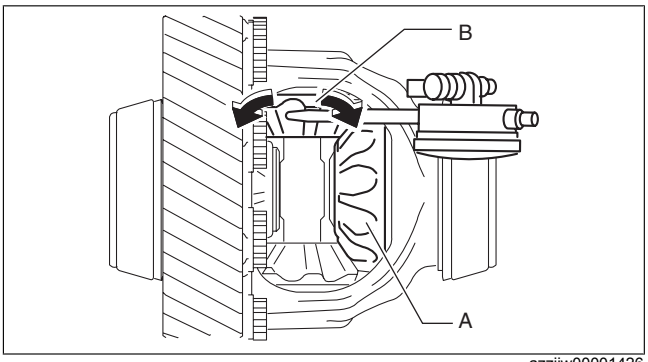
Caution
- **Because a difference occurs in the backlash measurement value if the secured side gear moves, move the pinion gear by hand so that the secured side gear does not move.**

cardiagn.com

A : Secure by hand
B : Move the pinion gear by hand so that the secured side gear does not move.

7. Input the measured backlash of the side gear on the front side and the pinion gear into the measurement/ adjustment value input sheet.

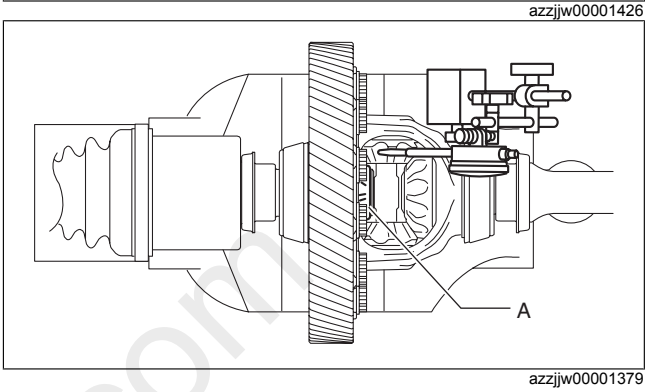
Note
- Input into section A in the measurement/ adjustment value input sheet.



azzjjw00001426

8. Secure the side gear on the rear side by hand.

A : Secure by hand



azzjjw00001379

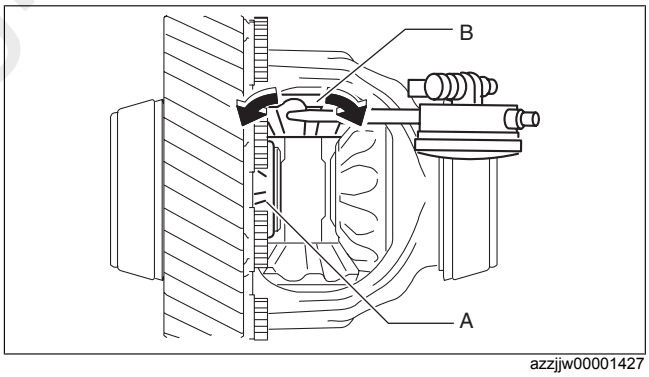
9. Move the pinion gear with the dial gauge by hand and measure the backlash.

Caution
- **Because a difference occurs in the backlash measurement value if the secured side gear moves, move the pinion gear by hand so that the secured side gear does not move.**

A : Secure by hand
B : Move the pinion gear by hand so that the secured side gear does not move.

10. Input the measured backlash of the side gear on the rear side and the pinion gear into the measurement/ adjustment value input sheet.

Note
- Input into section B in the measurement/ adjustment value input sheet.



azzjjw00001427

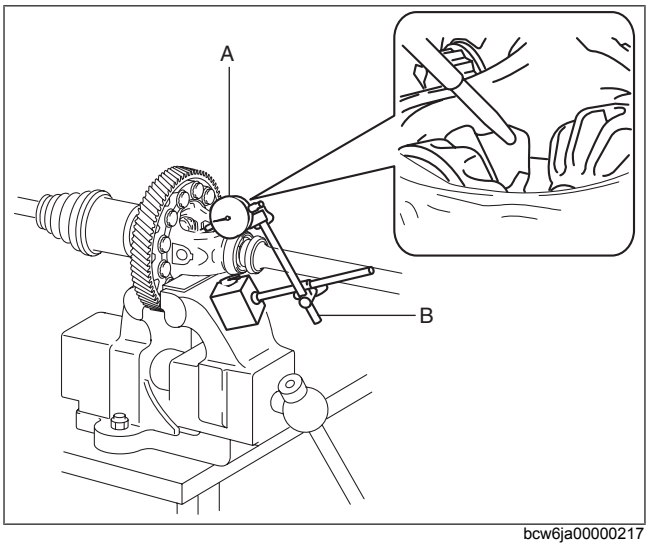
11. Set the dial gauge and magnetic stand to the pinion gear on the opposite side.

Caution
- **To reduce difference during the backlash measurement, set the dial gauge so that it is perpendicular to the teeth of the pinion gear.**

cardiagn.com

A : Dial gauge

B : Magnetic stand



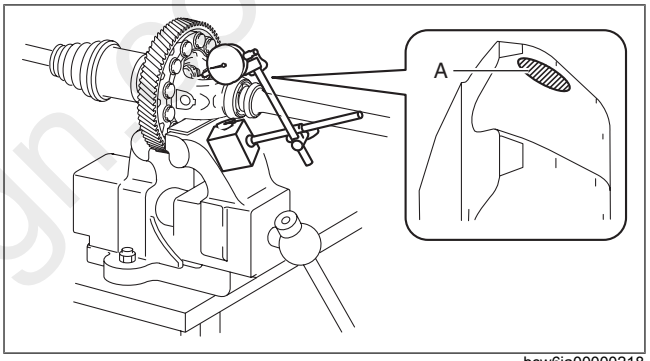
bcw6ja00000217

12. Set the dial gauge end to the pinion gear teeth.

Caution

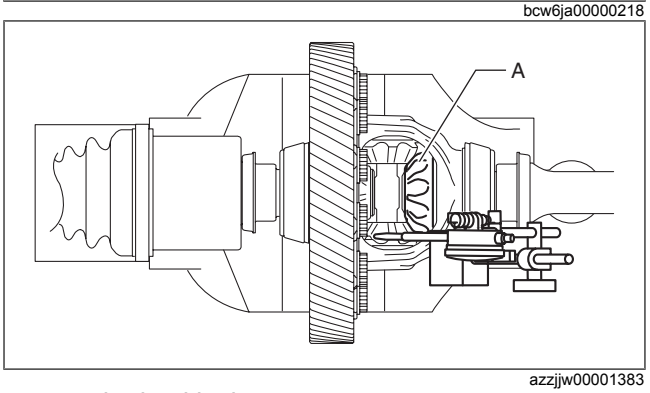
- **To reduce difference during the backlash measurement, set the dial gauge end within the area shown in the figure.**

A : Dial gauge end set area



bcw6ja00000218

13. Secure the side gear on the front side by hand.
A : Secure by hand



azzjjw00001383

14. Move the pinion gear with the dial gauge by hand and measure the backlash.

Caution

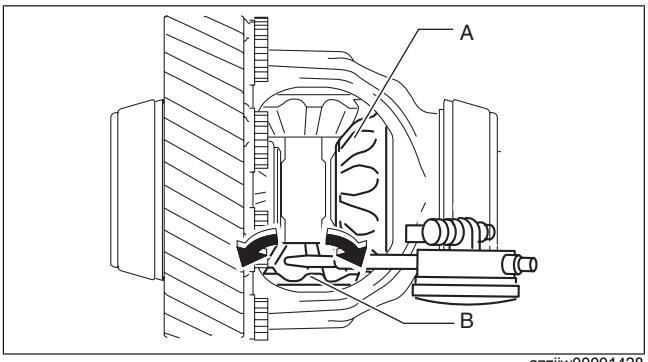
- **Because a difference occurs in the backlash measurement value if the secured side gear moves, move the pinion gear by hand so that the secured side gear does not move.**

cardiagn.com

A : Secure by hand
B : Move the pinion gear by hand so that the secured side gear does not move.

15. Input the measured backlash of the side gear on the front side and the pinion gear into the measurement/adjustment value input sheet.

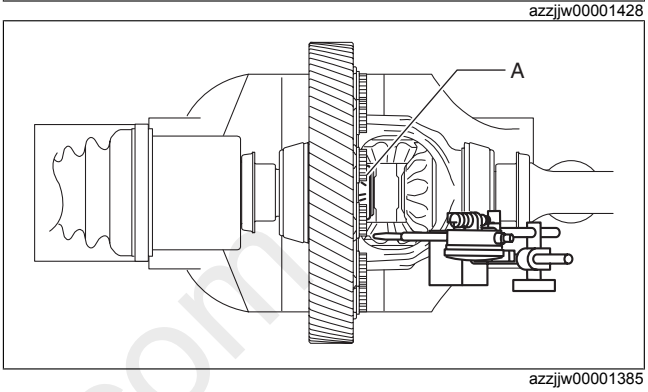
Note
- Input into section A in the measurement/adjustment value input sheet.



azzjjw00001428

16. Secure the side gear on the rear side by hand.

A : Secure by hand



azzjjw00001385

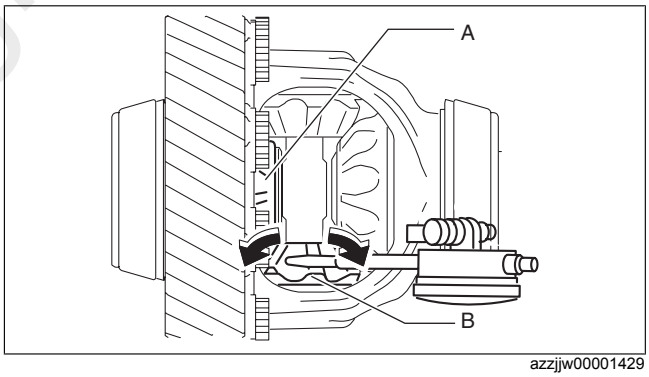
17. Move the pinion gear with the dial gauge by hand and measure the backlash.

Caution
- **Because a difference occurs in the backlash measurement value if the secured side gear moves, move the pinion gear by hand so that the secured side gear does not move.**

A : Secure by hand
B : Move the pinion gear by hand so that the secured side gear does not move.

18. Input the measured backlash of the side gear on the rear side and the pinion gear into the measurement/adjustment value input sheet.

Note
- Input into section B in the measurement/adjustment value input sheet.



azzjjw00001429

19. Calculate the average value of the measured backlash of the side gear on the front side and the pinion gear.
20. Input the calculated average backlash value of the side gear on the front side and the pinion gear into the measurement/adjustment value input sheet.

Note
- The calculated average backlash value of the side gear on the front side and the pinion gear is the front side differential backlash.

21. Calculate the average value of the measured backlash of the side gear on the rear side and the pinion gear.
22. Input the calculated average backlash value of the side gear on the rear side and the pinion gear into the measurement/adjustment value input sheet.

Note
- The calculated average backlash value of the side gear on the rear side and the pinion gear is the rear side differential backlash.

23. Verify that the front side differential backlash and the rear side differential backlash satisfy the specification.

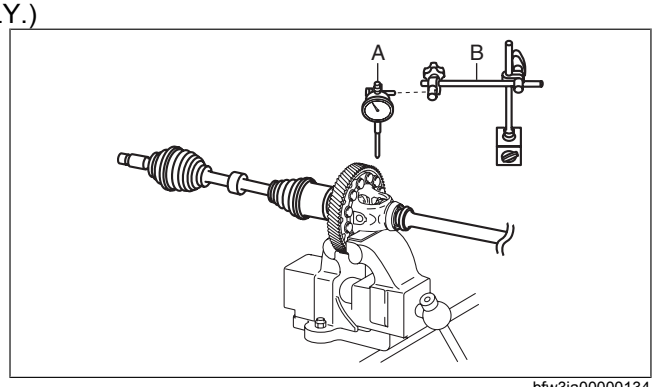
Specification
0.030—0.150 mm {0.0012—0.0059 in}

cardiagn.com

- If not within the specification, perform the differential backlash adjustment. (See Differential Backlash Adjustment.)
- After performing the differential backlash adjustment, if it exceeds the specification even with the thickest 0.95 mm {0.037 in} thrust washer assembled, disassemble the ring gear and differential and replace the differential gear case with a new one.
  (See RING GEAR AND DIFFERENTIAL DISASSEMBLY.)
  (See RING GEAR AND DIFFERENTIAL ASSEMBLY.)

24. Remove the dial gauge and magnetic stand.
A : Dial gauge

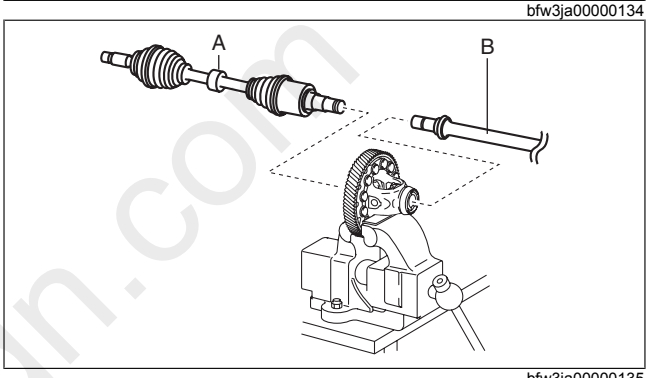
B : Magnetic stand



bfw3ja00000134

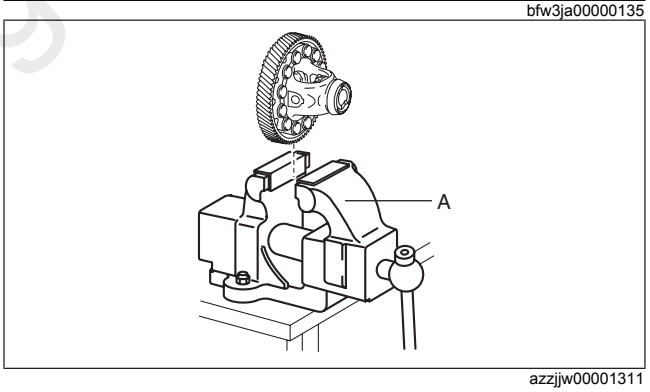
25. Remove the drive shaft.
A : Drive shaft (LH)

B : Drive shaft (RH)



bfw3ja00000135

26. Remove the ring gear and differential from the vise.
A : Vise



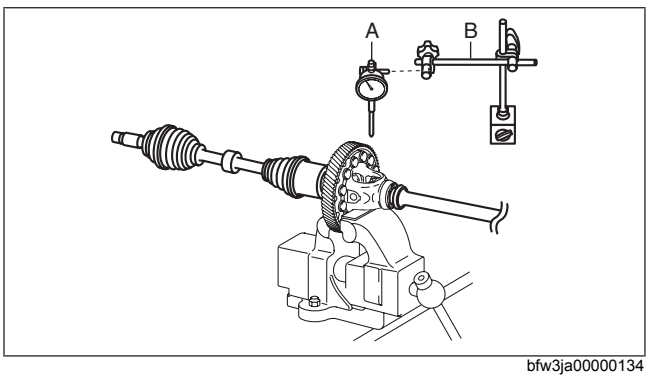
azzjjw00001311

Differential Backlash Adjustment

1. Remove the dial gauge and magnetic stand.

A : Dial gauge

B : Magnetic stand

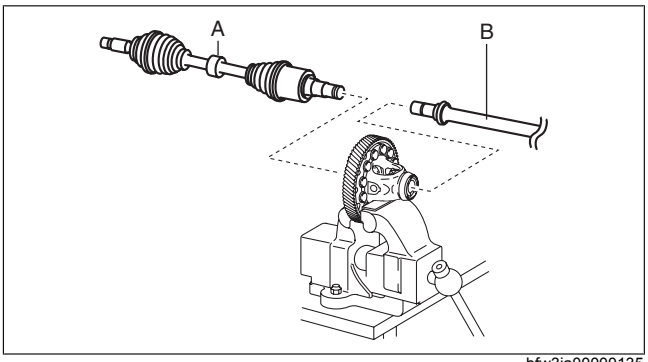


bfw3ja00000134

cardiagn.com

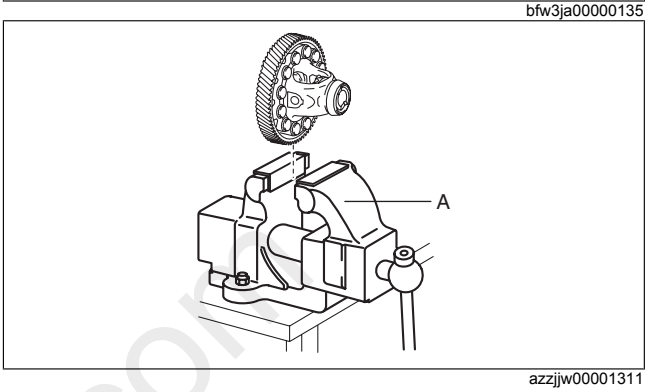
2. Remove the drive shaft.
A : Drive shaft (LH)

B : Drive shaft (RH)



bfw3ja00000135

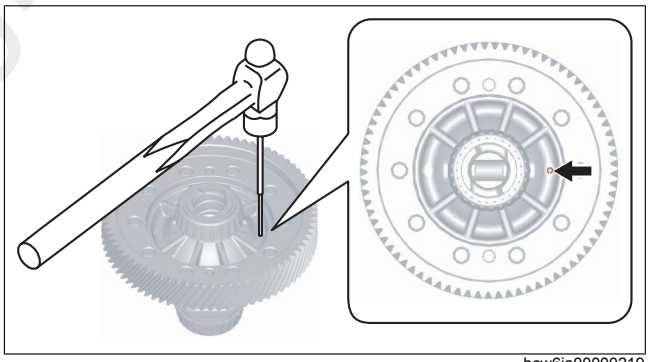
3. Remove the ring gear and differential from the vise.
A : Vise

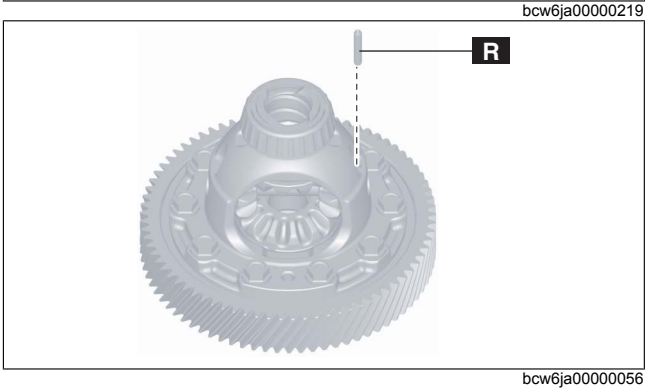


azzjjw00001311

4. Remove the roll pin shown in the figure using a pin punch.

Note
- Use a pin punch with an end outer diameter of 3 mm {0.1 in} or more and less than 4 mm {0.2 in}, and an end length of 50 mm {2.0 in} or more.

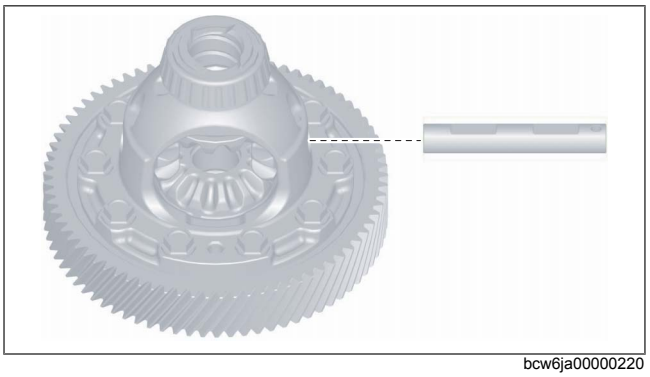

bcw6ja00000219



bcw6ja00000056

cardiagn.com

5. Remove the pinion shaft.



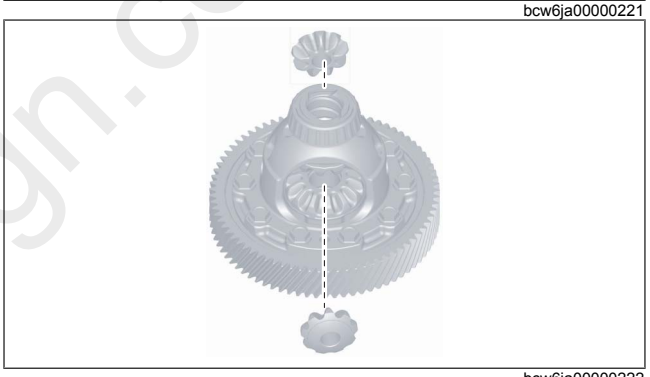
bcw6ja00000220

6. Remove the pinion gears using the following procedure:
   (1) Rotate the pinion gears as shown in the figure.



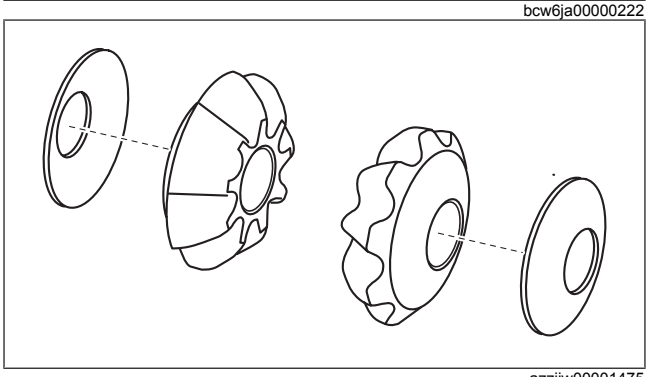
bcw6ja00000221

   (2) Remove the pinion gears.



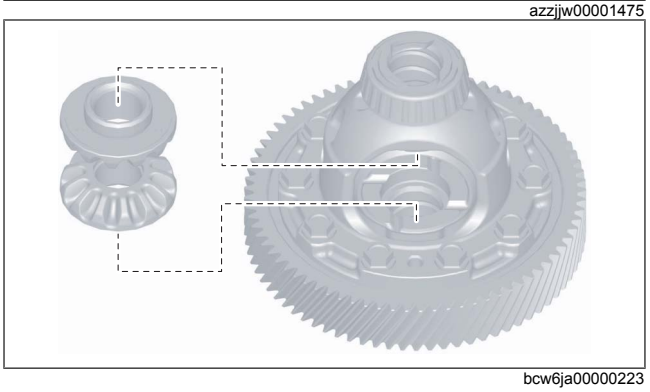
bcw6ja00000222

7. Remove the thrust washers from the pinion gears.



azzjjw00001475

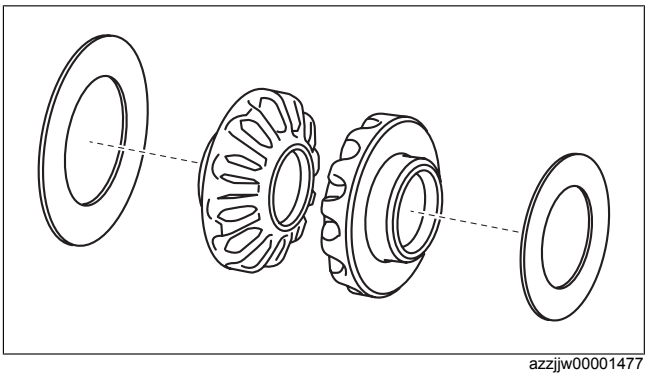
8. Remove the side gears.



bcw6ja00000223

cardiagn.com

9. Remove the thrust washers from the side gears.
10. Measure the thickness of the removed thrust washers from the side gears.


azzjjw00001477

Note
- Recommended measuring instrument: Micrometer

11. Input the measured thrust washer thickness into the measurement/adjustment value input sheet.
12. Perform the following calculation to calculate the differential backlash gap.

Note
- The differential backlash gap is the difference between the differential backlash and the median value of the differential backlash specification.

Front side differential backlash gap = C - J
C: Front side differential backlash
J: Median value of differential backlash specification (0.090 mm {0.0035 in})

Note
Example
C: Front side differential backlash is 0.165 mm {0.00650 in}
Front side differential backlash gap = 0.165 mm {0.00650 in} - 0.090 mm {0.0035 in} = 0.075 mm {0.0030 in}

Rear side differential backlash gap = D - J
D: Rear side differential backlash
J: Median value of differential backlash specification (0.090 mm {0.0035 in})

Note
Example
D: Rear side differential backlash is 0.155 mm {0.00610 in}
Rear side differential backlash gap = 0.155 mm {0.00610 in} - 0.090 mm {0.0035 in} = 0.065 mm {0.0026 in}

13. Input the differential backlash gap into the measurement/adjustment value input sheet.
14. Perform the following calculation to calculate the thrust washer thickness gap.

Note
- The thrust washer thickness gap is the difference between the removed thrust washer thickness and the optimum thrust washer thickness.
- If the thrust washer thickness is thickened 0.1 mm {0.004 in}, the differential backlash decreases approx. 0.08 mm {0.003 in}.

Front side thrust washer thickness gap = K × 0.1 mm {0.004 in} / 0.08 mm {0.003 in}
K: Front side differential backlash gap

Note
Example
K: Differential backlash gap on front side is 0.075 mm {0.0030 in}
Front side thrust washer thickness gap = 0.075 mm {0.0030 in} × 0.1 mm {0.004 in} / 0.08 mm {0.003 in} = 0.094 mm {0.0037 in}

Rear side thrust washer thickness gap = L × 0.1 mm {0.004 in} / 0.08 mm {0.003 in}
L: Rear side differential backlash gap

Note
Example
L: Differential backlash gap on rear side is 0.065 mm {0.0026 in}
Rear side thrust washer thickness gap = 0.065 mm {0.0026 in} × 0.1 mm {0.004 in} / 0.08 mm {0.003 in} = 0.081 mm {0.0032 in}

15. Input the calculated thrust washer thickness gap into the measurement/adjustment value input sheet.
16. Perform the following calculation to calculate the optimum thrust washer thickness.

carusa.com

Optimum front side thrust washer thickness = H + M
H: Thickness of removed front side thrust washer
M: Front side thrust washer thickness gap

Note
Example
H: Thickness of removed front side thrust washer is 0.810 mm {0.0032 in}
M: Front side thrust washer thickness gap is 0.094 mm {0.0037 in}
Optimum front side thrust washer thickness = 0.810 mm {0.0032 in} + 0.094 mm {0.0037 in} = 0.904 mm {0.0356 in}

Optimum rear side thrust washer thickness = I + N
I: Thickness of removed rear side thrust washer
N: Rear side thrust washer thickness gap

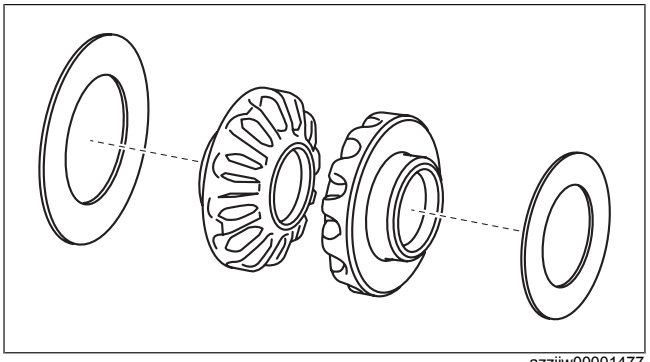
Note
Example
I: Thickness of removed thrust washer on rear side is 0.795 mm {0.0313 in}
N: Thrust washer thickness gap on rear side is 0.081 mm {0.0032 in}
Optimum rear side thrust washer thickness = 0.795 mm {0.0313 in} + 0.081 mm {0.0032 in} = 0.876 mm {0.0345 in}

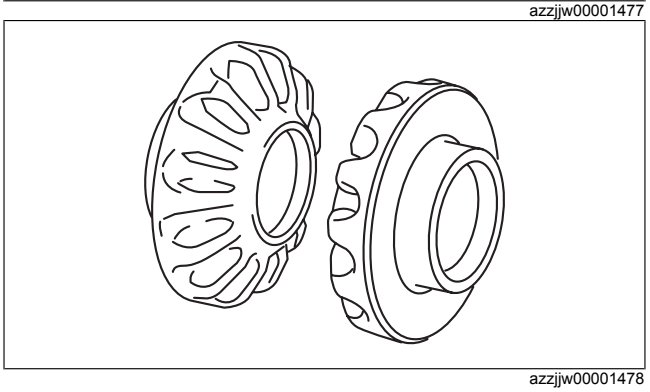
17. Input the calculated optimum thrust washer thickness into the measurement/adjustment value sheet.
18. From the following table, select a thrust washer of a thickness closest to the calculated optimum thrust washer thickness.

Caution
- **Select the same thickness of the thrust washers on the front and rear sides. Measure the differential backlash again with the thrust washers of the same thickness assembled on the front and rear sides, and if the measurement value does not satisfy the specification, thrust washers of different thickness can be used on the front and rear sides.**

Thrust washer thickness
0.95 mm {0.037 in}
0.90 mm {0.035 in}
0.85 mm {0.033 in}
0.80 mm {0.031 in}
0.75 mm {0.030 in}

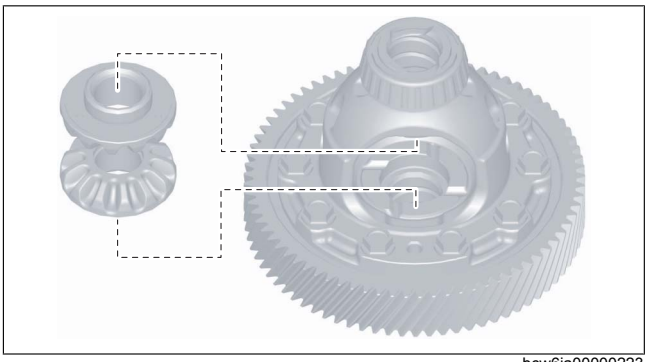
19. Assemble the selected thrust washers to the side gears.


azzjjw00001477


azzjjw00001478

cardiagn.com

20. Assemble the side gears which have the thrust washers assembled to them.

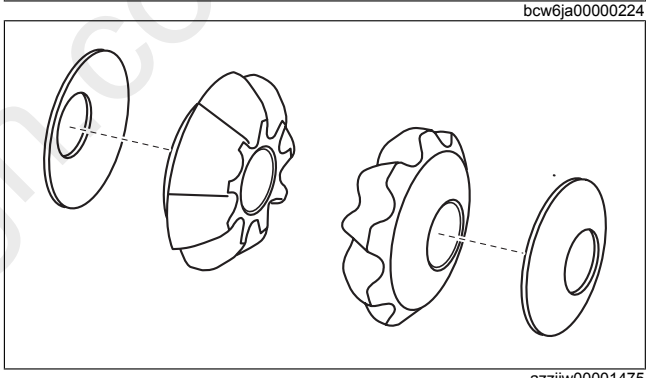


bcw6ja00000223

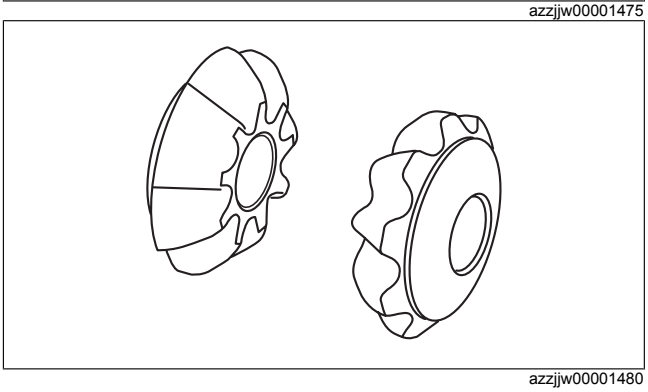


bcw6ja00000224

21. Assemble the thrust washers to the pinion gears.

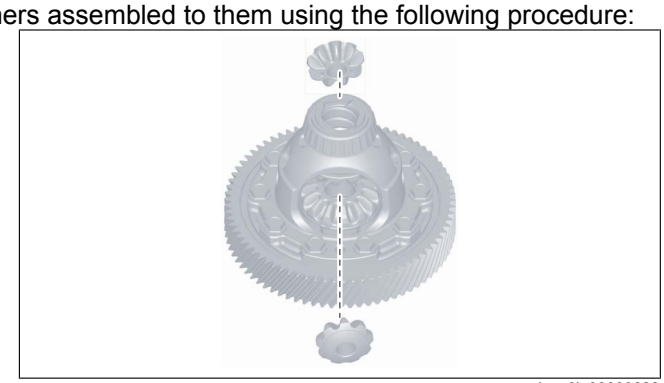


azzjjw00001475



azzjjw00001480

22. Assemble the pinion gears which have the thrust washers assembled to them using the following procedure:
    (1) Assemble the pinion gears which have the thrust washers assembled to them.



bcw6ja00000222

cardiagn.com



bcw6ja00000225

(2) Rotate the pinion gears so that the pinion shaft holes of the differential gear case and the pinion gears are aligned as shown in the figure.

Note

- If the pinion shaft holes of the differential gear case and the pinion gears are not aligned, remove the pinion gears, change the engagement with the side gears, and reassemble.

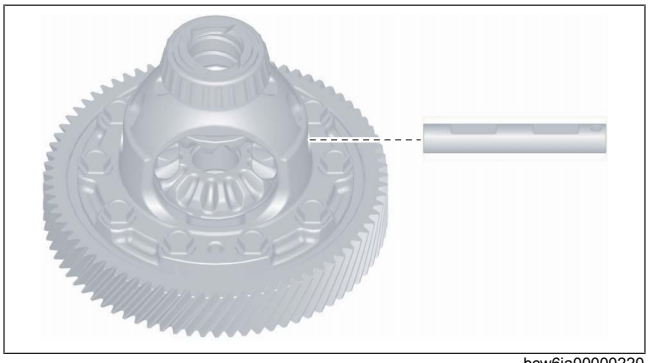


bcw6ja00000226

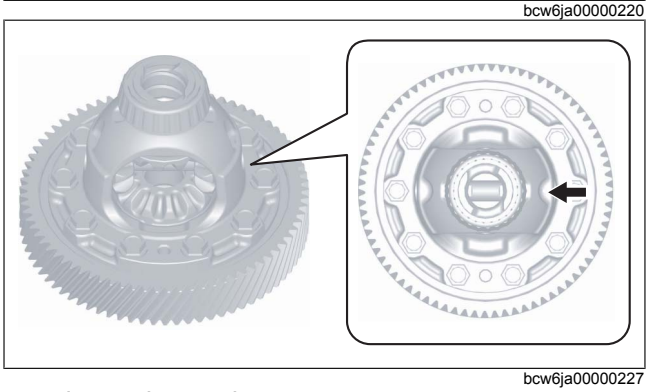
23. Assemble the pinion shaft.

Caution

- **Assemble the pinion shaft so that the roll pin holes of the differential gear case and the pinion shaft are aligned.**



bcw6ja00000220

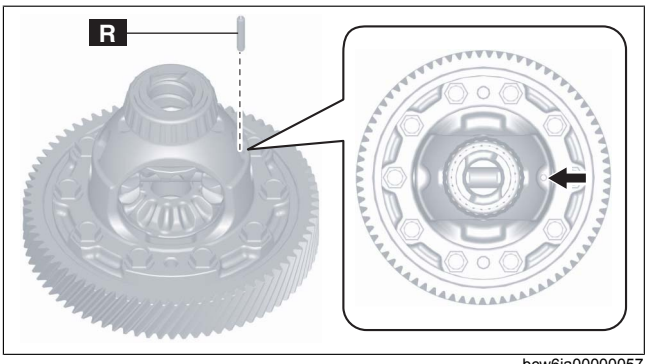


bcw6ja00000227

24. Assemble a new roll pin to the position shown in the figure using a pin punch.

cardiagn.com

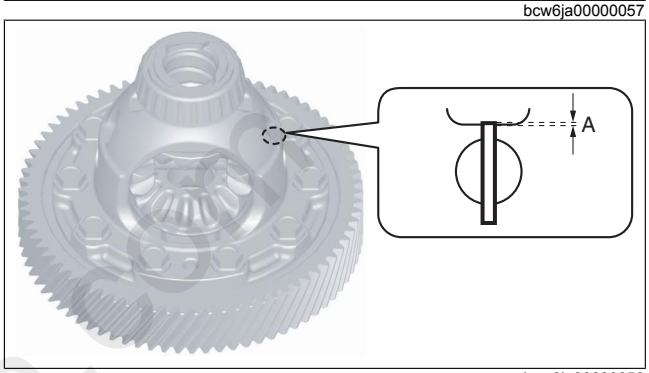
Note
- Use a pin punch with an end outer diameter of 5 mm {0.2 in} or more and less than 8 mm {0.3 in}.



bcw6ja00000057

A : -1—0 mm {-0.03—0 in}

25. Perform the differential backlash measurement. (See Differential Backlash Measurement.)



bcw6ja00000058

cardiagn.com